

Homework 6 Solutions

Based on the third edition of *Abstract Algebra* of Dummit and Foote.

Section 4.1: Group Actions and Permutation Representations

Section 1, Exercise 1

Let G act on the set A . Prove that if $a, b \in A$ and $b = g \cdot a$ for some $g \in G$, then $G_b = gG_ag^{-1}$ (G_a is the stabilizer of a). Deduce that if G acts transitively on A then the kernel of the action is $\bigcap_{g \in G} gG_ag^{-1}$.

Solution. We first compare the two stabilizers. For any $h \in G$,

$$\begin{aligned} h \in G_b &\iff h \cdot b = b \\ &\iff h \cdot (g \cdot a) = g \cdot a \\ &\iff (g^{-1}hg) \cdot a = a \\ &\iff g^{-1}hg \in G_a \\ &\iff h \in gG_ag^{-1}. \end{aligned}$$

Thus $G_b = gG_ag^{-1}$. The kernel of the action consists precisely of the elements fixing every point of A , so it is

$$\ker(G \curvearrowright A) = \bigcap_{b \in A} G_b.$$

If the action is transitive, every $b \in A$ has the form $b = g \cdot a$ for some $g \in G$. Using the stabilizer identity just proved, we obtain

$$\ker(G \curvearrowright A) = \bigcap_{b \in A} G_b = \bigcap_{g \in G} gG_ag^{-1}.$$

Section 4.2: Groups Acting on Themselves by Left Multiplication

Section 2, Exercise 8

Prove that if H has finite index n then there is a normal subgroup K of G with $K \leq H$ and $|G : K| \leq n!$.

Solution. Let G act by left multiplication on the set

$$\Omega = \{gH : g \in G\}$$

of left cosets of H . Since $|G : H| = n$, the set Ω has n elements, and the action gives a homomorphism

$$\pi_H : G \longrightarrow S_\Omega \cong S_n.$$

Set $K = \ker \pi_H$ and recall that it is a normal subgroup of G . As the kernel of a homomorphism, $K \trianglelefteq G$. Moreover, K fixes every coset, in particular the coset H ; its stabilizer is H . Hence $K \leq H$. By the First Isomorphism Theorem, $G/K \cong \pi_H(G) \leq S_n$ whence G/K is finite and

$$|G : K| = \text{ord } G/K = \text{ord } \pi_H(G) \leq \text{ord } S_n = n!.$$

Section 2, Exercise 10

Prove that every non-abelian group of order 6 has a nonnormal subgroup of order 2. Use this to classify groups of order 6. [Produce an injective homomorphism into S_3 .]

Solution. Let G be a non-abelian group of order 6. By Cauchy's Theorem, G has an element x of order 3 and an element y of order 2. Put $A = \langle x \rangle$ and $H = \langle y \rangle$. Since A has index 2, it is normal in G .

We claim that H is not normal. If it were, then A and H would both be normal, and $A \cap H = 1$ because their orders are relatively prime. For $a \in A$ and $h \in H$, the commutator $aha^{-1}h^{-1}$ lies in both A and H : normality of A places it in A , and normality of H places it in H . Hence the commutator is 1, so A and H commute elementwise. Since

$$\text{ord } AH = \frac{\text{ord } A \text{ ord } H}{\text{ord } A \cap H} = 6,$$

we would have $G = AH \cong A \times H$, which is abelian, a contradiction. Thus H is a nonnormal subgroup of order 2.

Now let G act by left multiplication on its three left cosets of H . This produces a homomorphism

$$\varphi : G \longrightarrow S_3.$$

Its kernel is a normal subgroup contained in H . Because $\text{ord } H = 2$, the kernel is either 1 or H . It cannot be H , since that would make H normal in G . Therefore $\ker \varphi = 1$, so φ is injective. Both G and S_3 have order 6, and hence $G \cong S_3$.

It remains to identify the abelian case. If G is abelian of order 6, Cauchy's Theorem gives commuting elements x, y of orders 3 and 2, respectively. Then xy has order 6, so $G = \langle xy \rangle \cong \mathbb{Z}/6\mathbb{Z}$. Consequently, up to isomorphism the groups of order 6 are exactly $\mathbb{Z}/6\mathbb{Z}$ and S_3 .

Section 2, Exercise 14

Let G be a finite group of composite order n with the property that G has a subgroup of order k for each positive integer k dividing n . Prove that G is not simple.

Solution. Let p be the smallest prime dividing n . Since n is composite, $1 < n/p < n$. By hypothesis, G has a subgroup H of order n/p . Its index is

$$|G : H| = \frac{n}{n/p} = p,$$

whence $H \trianglelefteq G$. Clearly it is a nontrivial proper normal subgroup of G , therefore G is not simple.

Section 4.3: Groups Acting on Themselves by Conjugation

Section 3, Exercise 4

Prove that if $S \leq G$ and $g \in G$ then $gN_G(S)g^{-1} = N_G(gSg^{-1})$ and $gC_G(S)g^{-1} = C_G(gSg^{-1})$.

Solution. We prove the normalizer identity first. Let $x \in gN_G(S)g^{-1}$, so $x = gng^{-1}$ for some $n \in N_G(S)$. Then

$$\begin{aligned} x(gSg^{-1})x^{-1} &= gng^{-1}(gSg^{-1})gn^{-1}g^{-1} \\ &= g(nSn^{-1})g^{-1} \\ &= gSg^{-1}. \end{aligned}$$

Thus $x \in N_G(gSg^{-1})$, proving one inclusion. Replacing g by g^{-1} and S by gSg^{-1} gives the reverse inclusion. Hence $gN_G(S)g^{-1} = N_G(gSg^{-1})$.

For centralizers, let $x = gcg^{-1}$ with $c \in C_G(S)$. For every $s \in S$,

$$x(gsg^{-1})x^{-1} = g(csc^{-1})g^{-1} = gsg^{-1}.$$

Therefore x centralizes every element of gSg^{-1} , so $gC_G(S)g^{-1} \leq C_G(gSg^{-1})$. Applying the same argument as before gives the reverse inclusion.

Section 3, Exercise 5

If the center of G is of index n , prove that every conjugacy class has at most n elements.

Solution. Fix $x \in G$. Its conjugacy class has cardinality

$$\text{ord cl}_G(x) = |G : C_G(x)|$$

by the orbit-stabilizer theorem for the conjugation action. Every central element commutes with x , so $Z(G) \leq C_G(x)$. Consequently

$$\text{ord cl}_G(x) = |G : C_G(x)| \leq |G : Z(G)| \leq n.$$

Thus every conjugacy class contains at most n elements.

Section 3, Exercise 6

Assume G is a non-abelian group of order 15. Prove that $Z(G) = 1$. Use the fact that $\langle g \rangle \leq C_G(g)$ for all $g \in G$ to show that there is at most one possible class equation for G . [Use Exercise 36, Section 3.1. (solved in [Homework 4](#))]

Solution. Since $\text{ord } G = 3 \cdot 5$, Lagrange's Theorem shows that a nontrivial proper center would have order 3 or 5. In that case $G/Z(G)$ would have prime order, hence would be cyclic. Exercise 36 of Section 3.1 states that if $G/Z(G)$ is cyclic, then G is abelian, contrary to the hypothesis. Also $Z(G) \neq G$ because G is non-abelian. Therefore $Z(G) = 1$.

Let $g \neq 1$. Its order divides 15, but it cannot be 15, since then G would be cyclic. Thus $\text{ord } g$ is 3 or 5. Because $\langle g \rangle \leq C_G(g)$, the order of $C_G(g)$ is divisible by $\text{ord } g$. It is also a divisor of 15. If $C_G(g) = G$, then $g \in Z(G)$, which is impossible. Consequently,

$$\text{ord } C_G(g) = \begin{cases} 3, & \text{ord } g = 3, \\ 5, & \text{ord } g = 5. \end{cases}$$

The conjugacy class of an element of order 3 therefore has size $15/3 = 5$, while the class of an element of order 5 has size $15/5 = 3$.

If there are a noncentral classes of size 5 and b noncentral classes of size 3, the class equation must be

$$15 = 1 + 5a + 3b.$$

It has at most one possible solution, namely

$$\boxed{15 = 1 + 5 + 3 + 3 + 3}.$$

Section 3, Exercise 23

Recall (cf. Exercise 16, Section 2.4) that a proper subgroup M of G is called maximal if whenever $M \leq H \leq G$, either $H = M$ or $H = G$. Prove that if M is a maximal subgroup of G then either $N_G(M) = M$ or $N_G(M) = G$. Deduce that if M is a maximal subgroup of G that is not normal in G then the number of nonidentity elements of G that are contained in conjugates of M is at most $(\text{ord } M - 1)|G : M|$.

Solution. Every subgroup normalizes itself, so

$$M \leq N_G(M) \leq G.$$

Since M is maximal, there is no subgroup strictly between M and G . Therefore either $N_G(M) = M$ or $N_G(M) = G$.

Now assume that M is not normal in G . Then $N_G(M) \neq G$, so the first part forces $N_G(M) = M$. Under the conjugation action of G on its subgroups, the stabilizer of M is $N_G(M)$. Orbit-stabilizer therefore shows that the number of distinct conjugates of M is

$$|G : N_G(M)| = |G : M|.$$

Each conjugate contains exactly $\text{ord } M - 1$ nonidentity elements. The sets of such elements may overlap, so the size of their union is no larger than the sum of their sizes. Hence the number of nonidentity elements of G lying in conjugates of M is at most $(\text{ord } M - 1)|G : M|$.

Section 3, Exercise 27

Let $g_1, g_2, \dots, g_r$ be representatives of the conjugacy classes of the finite group G and assume these elements pairwise commute. Prove that G is abelian.

Solution. Fix $i \in \{1, \dots, r\}$ and set $C_i = C_G(g_i)$. Since the chosen representatives commute pairwise, every g_j belongs to C_i . Every element of G is conjugate to one of the representatives g_j ; hence every element of G belongs to some conjugate of C_i . Thus

$$G = \bigcup_{x \in G} xC_i x^{-1}.$$

Exercise 24 of this section says that a finite group cannot be the union of the conjugates of a proper subgroup. For completeness, this consequence also follows quickly from Exercise 23. If C_i were proper, place it in a maximal subgroup M . Then G would be the union of the conjugates of M . If $M \trianglelefteq G$, that union is just $M \neq G$. If M is not normal, Exercise 23 bounds the number of nonidentity elements in its conjugates by

$$(\text{ord } M - 1)|G : M| = \text{ord } G - |G : M| < \text{ord } G - 1,$$

so even after including the identity their union still has fewer than $\text{ord } G$ elements. Both cases are impossible. Hence $C_i = G$, and therefore $g_i \in Z(G)$.

This holds for every representative g_i . Finally, an element conjugate to a central element is that central element itself. Since every element of G is conjugate to one of the g_i , every element of G lies in $Z(G)$. Hence $G = Z(G)$, so G is abelian.